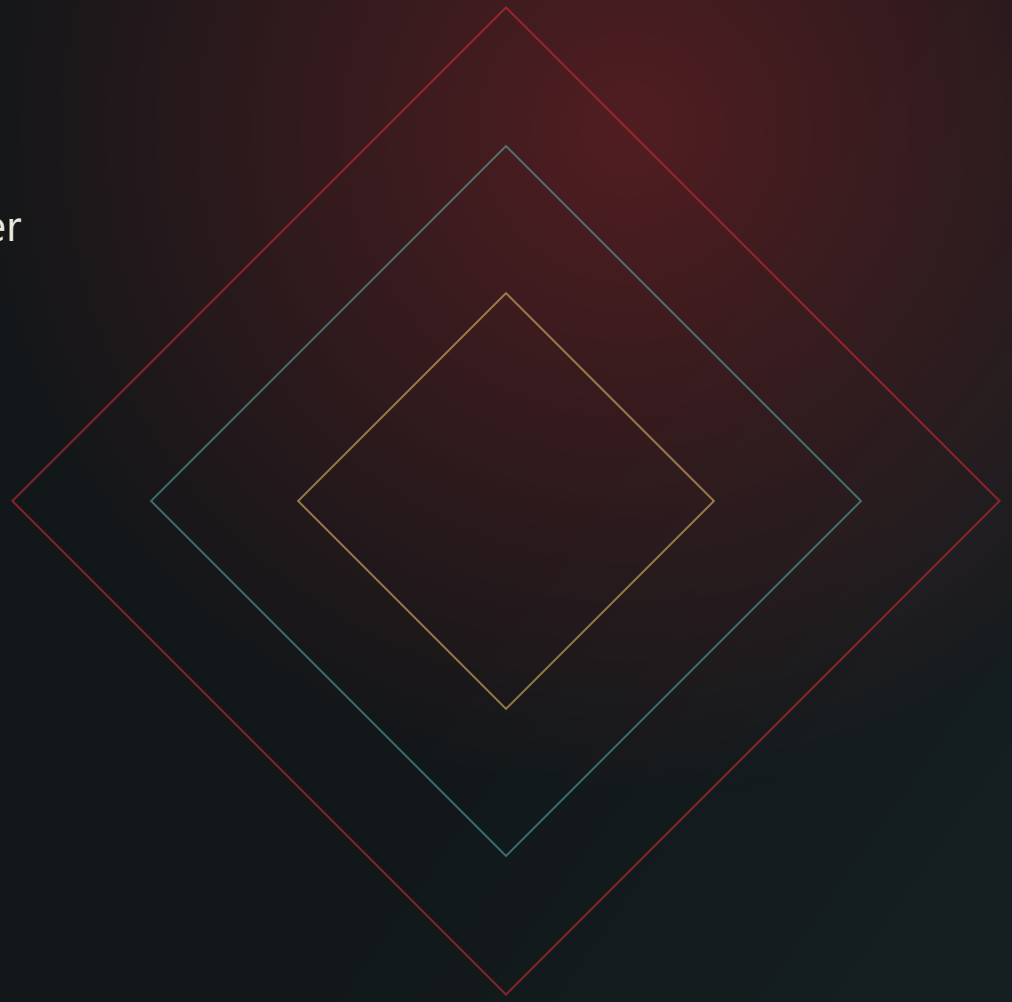
AMD RADEON HACKATHON 2026 / TRACK 2

EduCopilot

A private, agentic teaching assistant that turns course evidence into better teaching decisions.

Local AI | ROCm-ready | OpenAI-compatible | Outcome-based education



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Teaching is a connected workflow. Most tools are not.

> “What happened in the classroom today should improve what happens in the next class.”

Instructors repeatedly translate the same evidence across syllabi, spreadsheets, assessments, lesson plans, slide decks, and accreditation files. EduCopilot makes those artifacts part of one persistent course memory.

The cost of fragmentation

- Planning is detached from actual delivery.
- Student scores are detached from teaching decisions.
- Generated content does not remember classroom feedback.
- Accreditation evidence is assembled at the end, manually.

Target user: an instructor managing a course from onboarding through continuous improvement.

02 / THE SOLUTION

One course memory.
Six specialized agents.

01 / PLAN

Syllabus → schedule

Extract COs, validate Bloom's verbs, plan every class.

›

02 / TRACK

Class → evidence

Log topics, questions, pace, and delivery status.

›

03 / CREATE

Evidence → content

Generate slides, handouts, and teacher notes.

›

04 / ADAPT

Scores → action

Profile cohorts and generate targeted exams.

›

05 / IMPROVE

Attainment → report

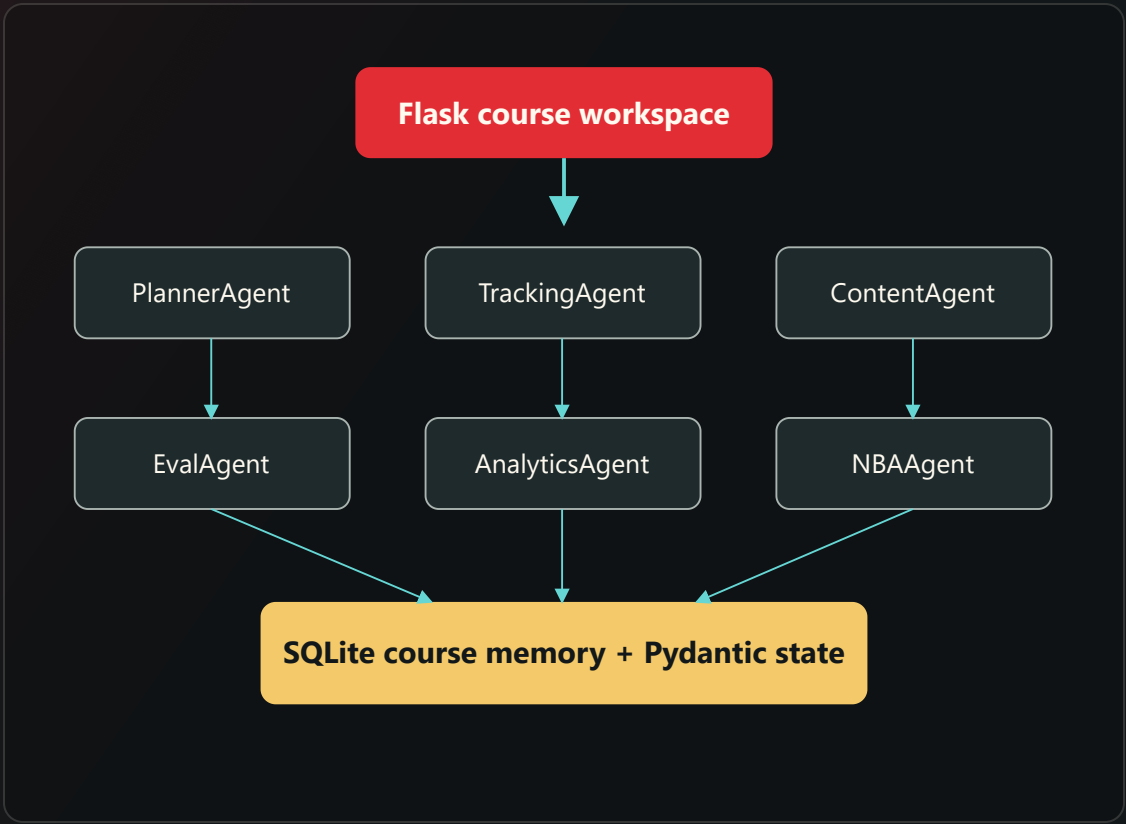
Compile CO evidence and next actions.

1
PERSISTENT COURSE WORKSPACE

6
TASK-SPECIALIZED AGENTS

∞
FEEDBACK LOOPS ACROSS A TERM

Reasoning is grounded in state, not just prompts.



Agentic loop

Retrieve relevant course state → reason with the model → validate structured output → persist the result → expose the next action in the UI.

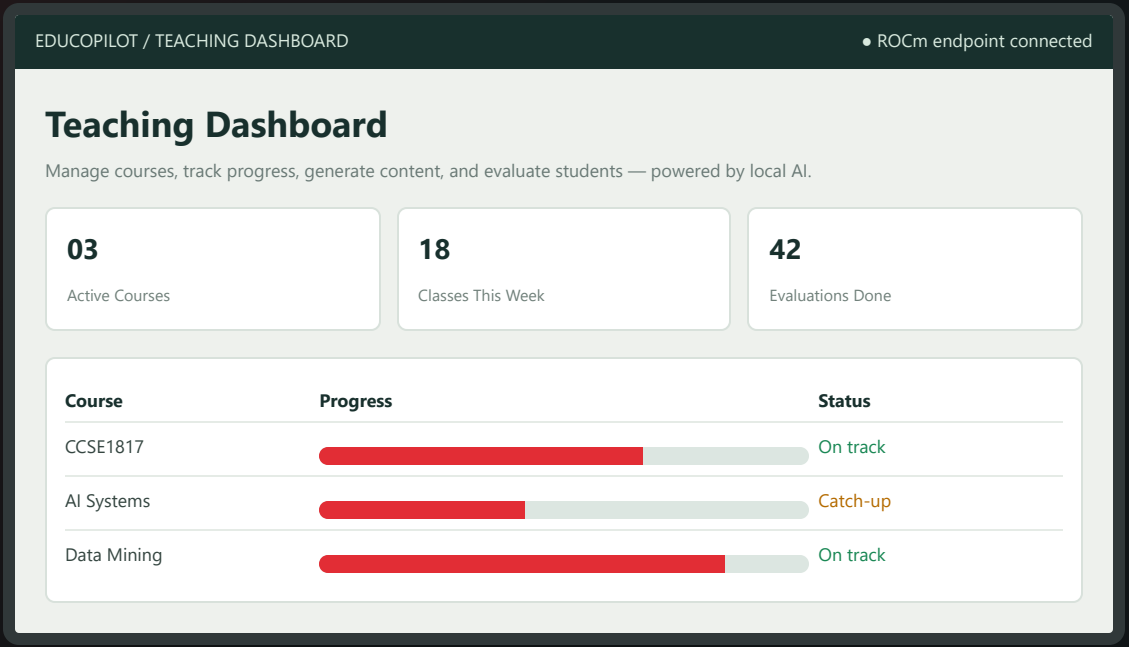
Private by design

Course materials, class feedback, scores, generated content, exams, and reports remain in the local application boundary.

Composable inference

The application speaks the OpenAI-compatible API. ROCm is the local acceleration layer behind the endpoint.

A teacher sees decisions, not infrastructure.



USER EXPERIENCE PRINCIPLES

Single course workspace

Planning, progress, content, assessments, exams, and accreditation live together.

Visible state

On-track status, cohort distribution, attainment gaps, and next actions are readable at a glance.

Human control

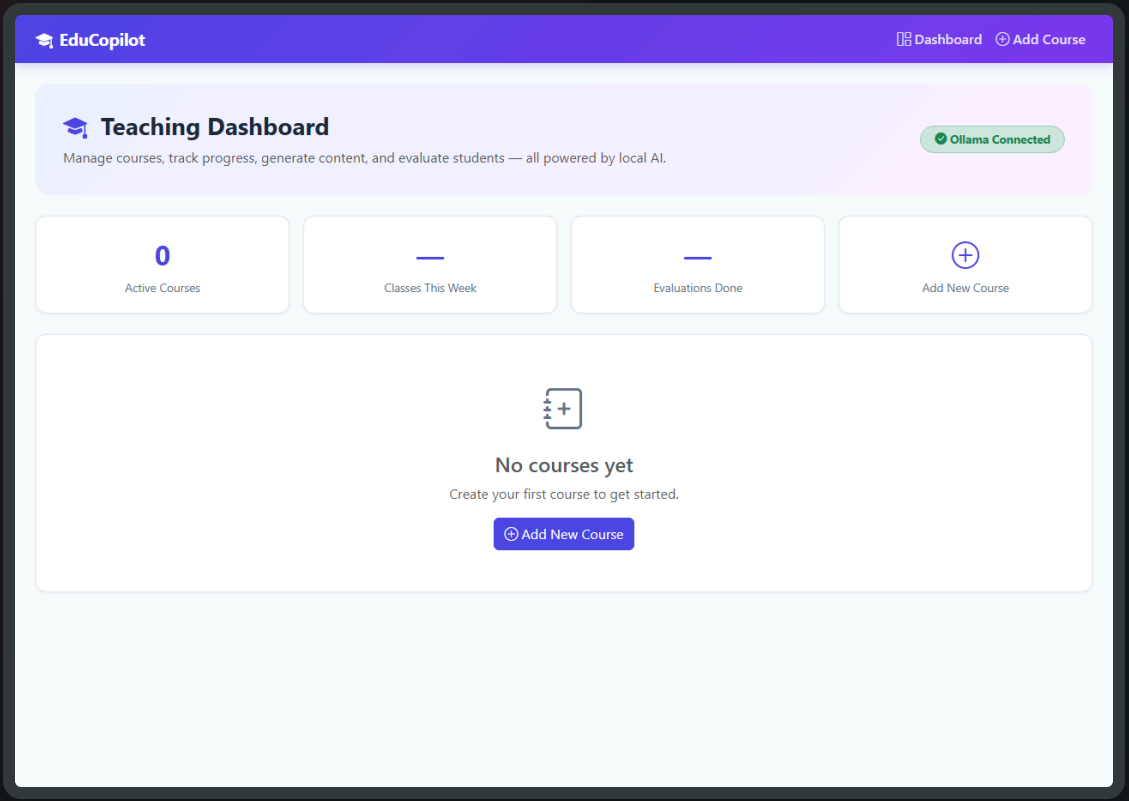
The instructor chooses the target lecture, cohort, CO, rubric, and final action.

Portable artifacts

Markdown reports, generated content, and PDF slides can leave the application.

INTERFACE VISUAL: REPRESENTATIVE PRODUCT VIEW BASED ON THE LIVE FLASK DASHBOARD

The running application keeps the human in the loop.



Captured from the local Flask application at `http://127.0.0.1:7860/`.

Visible connection state

The dashboard surfaces whether the configured OpenAI-compatible inference endpoint is reachable before the instructor starts a generation workflow.

Action-oriented UI

Courses, progress, content generation, evaluation, profiling, exams, and accreditation are presented as one navigable workspace.

Evidence-led workflow

The deck combines the real product capture with architecture diagrams and the completed local deployment model.

From syllabus to teachable plan in one action.

```
INPUT

Course: Computer Networks
Classes: 30
Teacher instruction: Include labs and case studies

OUTPUT

CO1 — Explain layered network models
Bloom: Understand / Valid

Lecture 07 — Routing algorithms
Target: CO2
Pedagogy: Case Study

Lecture 08 — Distance-vector routing
Target: CO2
Pedagogy: Lab
```

01

Extract

Course code, title, units, outcomes, and teaching constraints.

02

Audit

Map action verbs to Bloom's Taxonomy and flag weak outcomes.

03

Schedule

Distribute topics across the requested number of classes and attach pedagogy.

04

Remember

Persist the plan so every later agent can use it as course memory.

Classroom feedback becomes the next lesson's context.

01

Log the reality

Topics covered, student questions, teacher comments, and pace notes.

`SQLite log` `Lecture 07`

02

Detect the gap

Compare the actual topic with the planned topic and mark the delivery on track or behind.

`On track` `Catch-up`

03

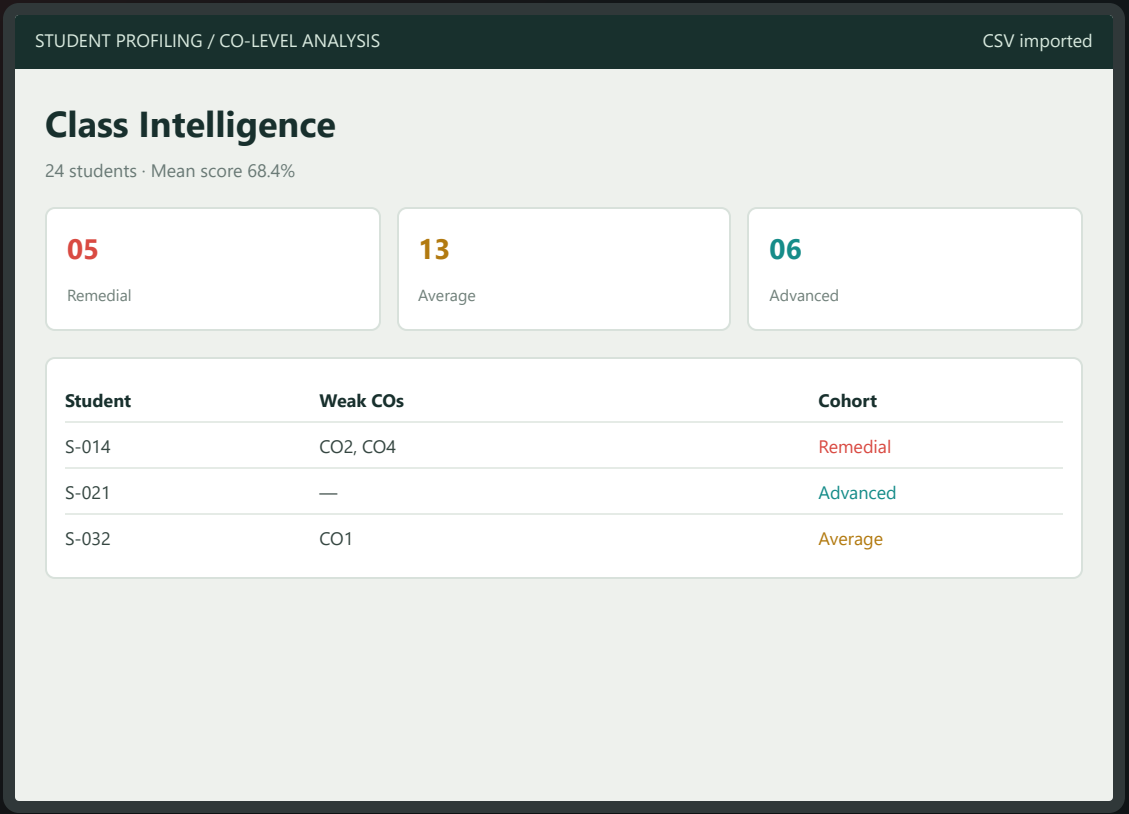
Adapt the response

Generate a compressed delivery plan or new content that addresses the observed need.

`Slides` `Teacher notes`

> Persistent memory turns a one-off completion into an evolving course relationship.

Same course. Different next steps.



Deterministic first

Pandas computes scores by student and Course Outcome. Thresholds produce reproducible cohorts.

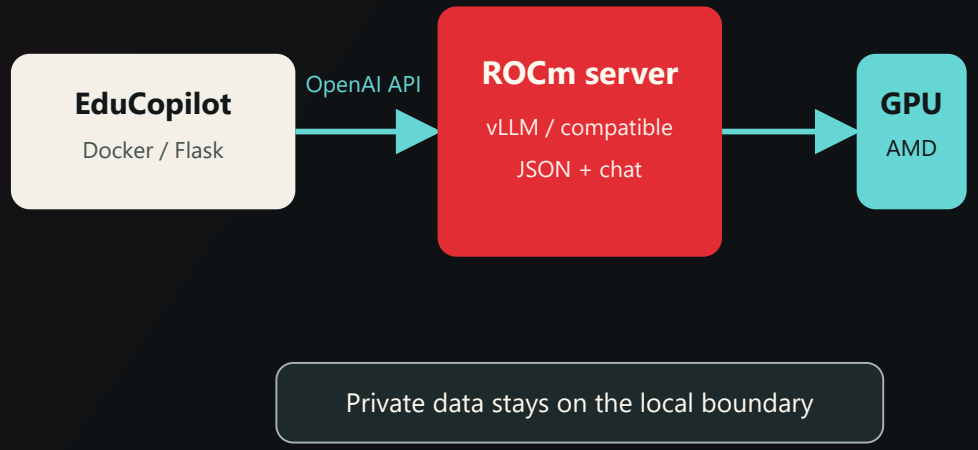
Reasoning second

The model turns the evidence into class health, weak topics, at-risk students, and recommended teacher actions.

Action third

Generate a Remedial, Average, or Advanced exam targeted to a selected CO and Bloom's distribution.

ROCm is the local inference engine behind the agent loop.




```
OPENAI_BASE_URL=http://host.docker.internal:8000/v1
OPENAI_API_KEY=your-api-key
OPENAI_MODEL=your-model-name

docker compose up --build -d
```

Why this boundary matters

- Swap model servers without rewriting agents.
- Use ROCm-enabled local inference for private data.
- Retain Ollama as a zero-config fallback for testing.

From clean checkout to private deployment in three commands.

```
# Configure ROCm endpoint
Copy-Item .env.example .env

# Edit .env with:
OPENAI_BASE_URL=http://host.docker.internal:8000/v1
OPENAI_API_KEY=********
OPENAI_MODEL=your-model

# Build and run
docker compose up --build -d

# Open
http://localhost:7860
```

Default fallback

If no `.env` exists, Compose uses local Ollama at **host.docker.internal:11434/v1** with **gemma4**.

Persistent local state

SQLite and Chroma data use the `educopilot_data` Docker volume so course state survives restarts.

One human control point

The instructor reviews generated plans, marks, questions, and accreditation mappings before use.

Built for useful teaching decisions and local acceleration.

Application value

- End-to-end instructor workflow
- Specialized agents and persistent memory
- Adaptive, outcome-based learning support
- Exportable content and accreditation evidence

Technical foundation

- Local ROCm model server
- Private inference boundary
- OpenAI-compatible integration
- Provider-neutral agent workflows

A useful private agent is not only a model call. It is a repeatable decision workflow with memory, tools, human review, and measurable execution.

Credible because the limits are visible.

Current boundaries

- Answer evaluation currently accepts pasted text rather than OCR or vision input.
- SQLite and single-user deployment are intended for local instructor use.
- Pacing detection uses topic-word overlap.
- CO-PO mapping requires institutional customization.

Responsible use

- Instructor reviews generated marks and exams.
- Do not send sensitive data to untrusted endpoints.
- Keep keys in `.env`, never in source control.
- Validate reports before official accreditation use.

Operational scope

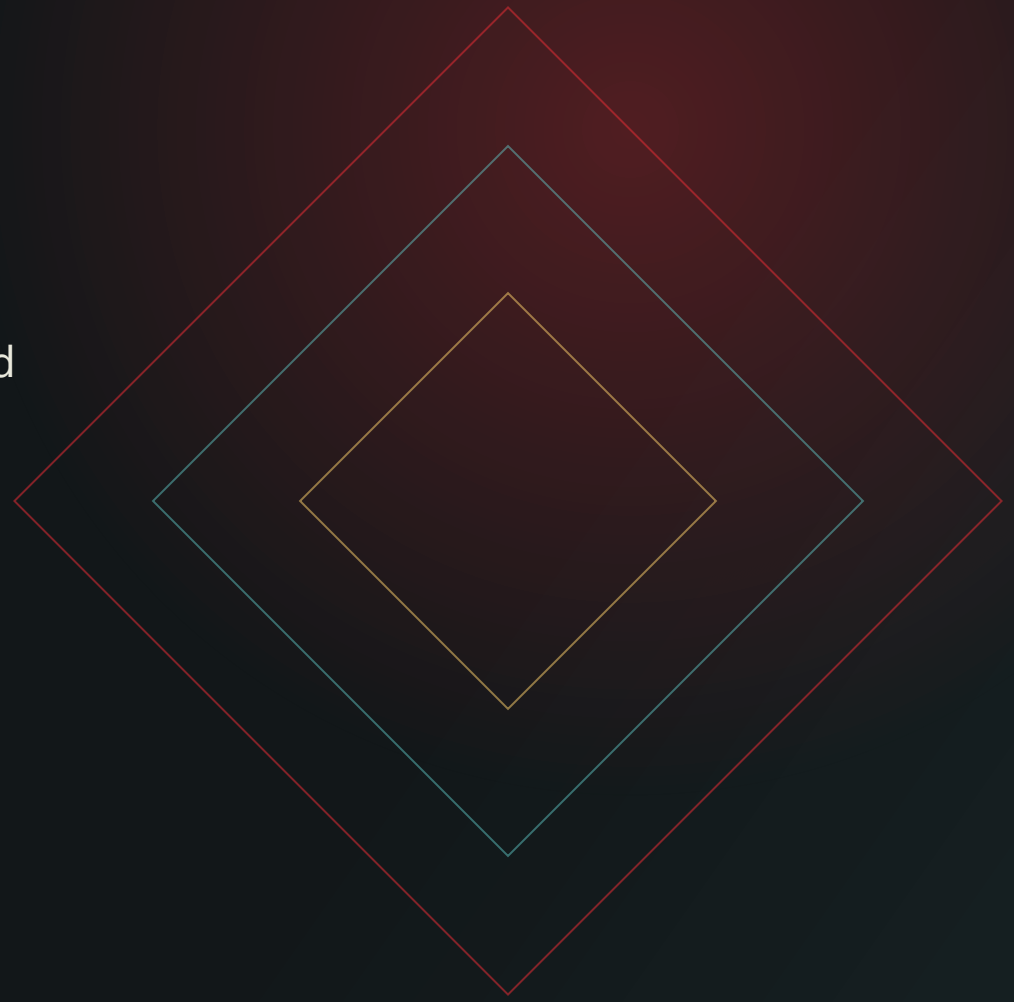
- Designed for a local instructor workspace.
- SQLite persistence supports course memory.
- OpenAI-compatible endpoints keep inference replaceable.
- Generated outputs remain reviewable and editable.

EduCopilot connects course operations today while keeping instructors responsible for final academic decisions.

THANK YOU

Make every class count.

EduCopilot connects planning, classroom evidence, adaptive support, and continuous improvement through private local AI.



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